Doubly Robust Estimation of Counterfactual Policy Mean Embeddings

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Outline

1. Motivations
2. Counterfactual Policy Mean Embeddings
3. A plug-in estimator
4. An efficient influence function based estimator
5. Applications

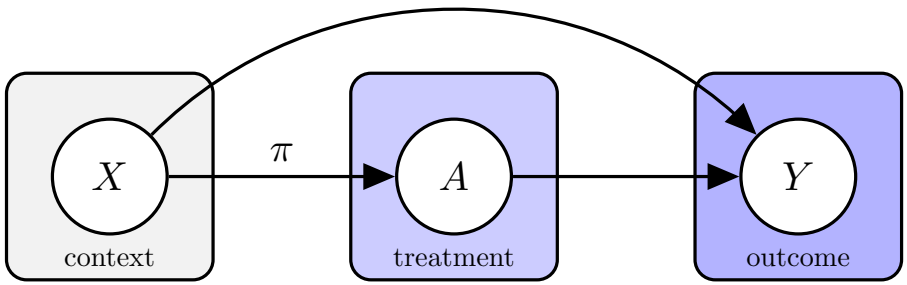
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Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, for each interaction the system:

- observed a context $x_i \in \mathcal{X}$, and $x_i \sim P_X$,
- sampled an action (treatment) $a_i \in \mathcal{A}$ from a logging policy π_0 with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,
- observes a loss (outcome) $y_i \sim P_Y(\cdot|x_i, a_i)$

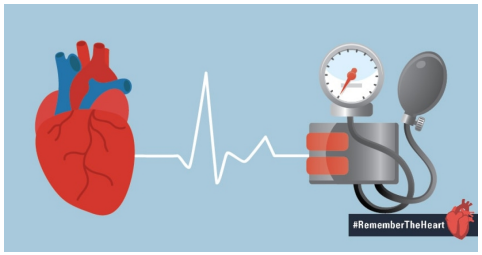


Counterfactual question: *"What would have been the outcome of the experiment, should we treat with stochastic intervention π'?"*

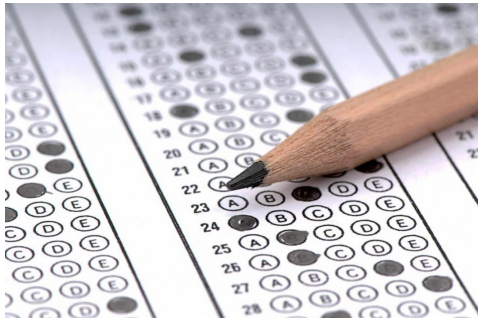
Outcomes

Traditionally, causal inference and policy evaluation target **scalar outcomes**:

- Health: blood pressure, cholesterol level, survival time.
- Economics: income, employment status, GDP growth.
- Education: test scores, years of schooling.







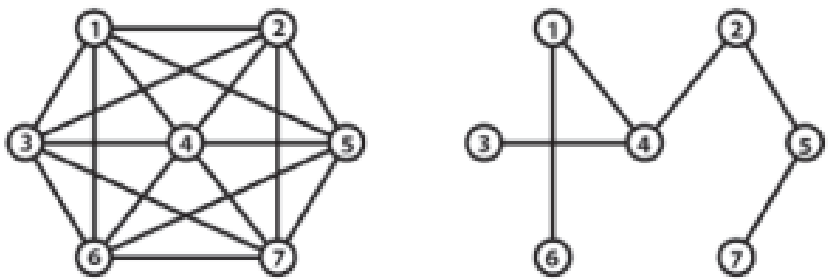
Structured Outcomes: Medical Imaging

- Treatment/Policy: choice of treatment (choice of drug, dosage of the drug).
- Outcome: **follow-up MRI/CT scan**.
- Goal: compare counterfactual distributions of lesion shapes, sizes, textures.



Structured Outcomes: Graphs

- Treatment/Policy: recommendation algorithm (who to follow, what to read).
- Outcome: **user's social network graph**.
- Question: do policies shift network topology (e.g., dense, sparse)?



Dense Sparse

Structured Outcomes: Text

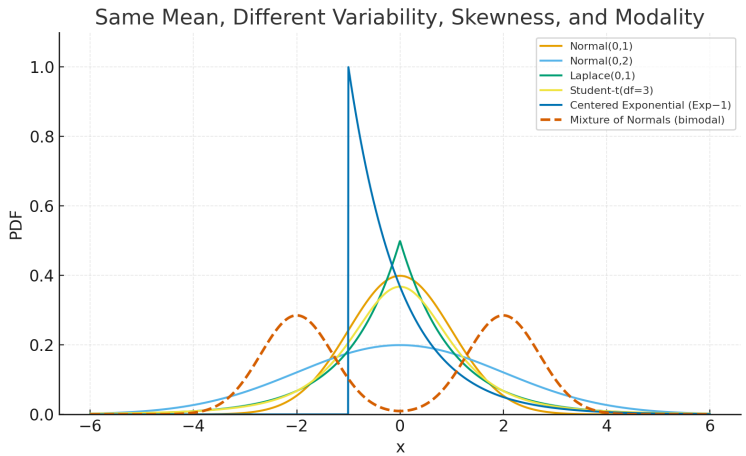
- Treatment/Policy: which message / text the system shows to the user.
- Outcome: **the user's written reply**.
- Goal: evaluate how alternative policies shift the distribution of replies (tone, length, engagement, sentiment).

Policy π (Neutral messages)
Prompt: *"Would you like to hear more about healthy diets?"*
User reply: *"Maybe, I'm not sure."*

Policy π' (Engaging messages)
Prompt: *"Discover how small changes in your diet can boost your energy!"*
User reply: *"Sounds interesting, tell me more!"*

Why look beyond counterfactual expectations?

- Classical causal inference tackles the **expected value function**: $\mathbb{E}[Y(\pi)]$.
- For **scalar outcomes**, this **expectation** often suffices (e.g. mean blood pressure reduction).
- However, for **structured outcomes** (e.g. images, text, graphs), averages are meaningless:
  - There is no such thing as an "average image" or "average essay".
- Furthermore, for all type of outcomes, the **entire outcome distribution** matters:
  - A policy may not change the mean outcome distribution, but it may **reduce variability** (risk reduction).
  - A policy may alter the **shape or tails** of the distribution (e.g. rare adverse events).



- We need tools to compare the **entire counterfactual distributions**.

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Counterfactual Policy Mean Embedding (CPME)

- Let $Y(a)$ denote the potential outcome under treatment a. A kernel $k_{\mathcal{Y}} : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ defines an RKHS $\mathcal{H}_{\mathcal{Y}}$.
- **Counterfactual Policy Mean Embeddings**[1] embed the distribution of $\{Y(a)\}_{a\sim\pi}$:

$$\chi(\pi) := \mathbb{E}_{P_Y \times \pi \times P_X}[k_{\mathcal{Y}}(\cdot, Y(a))] \in \mathcal{H}_{\mathcal{Y}}.$$

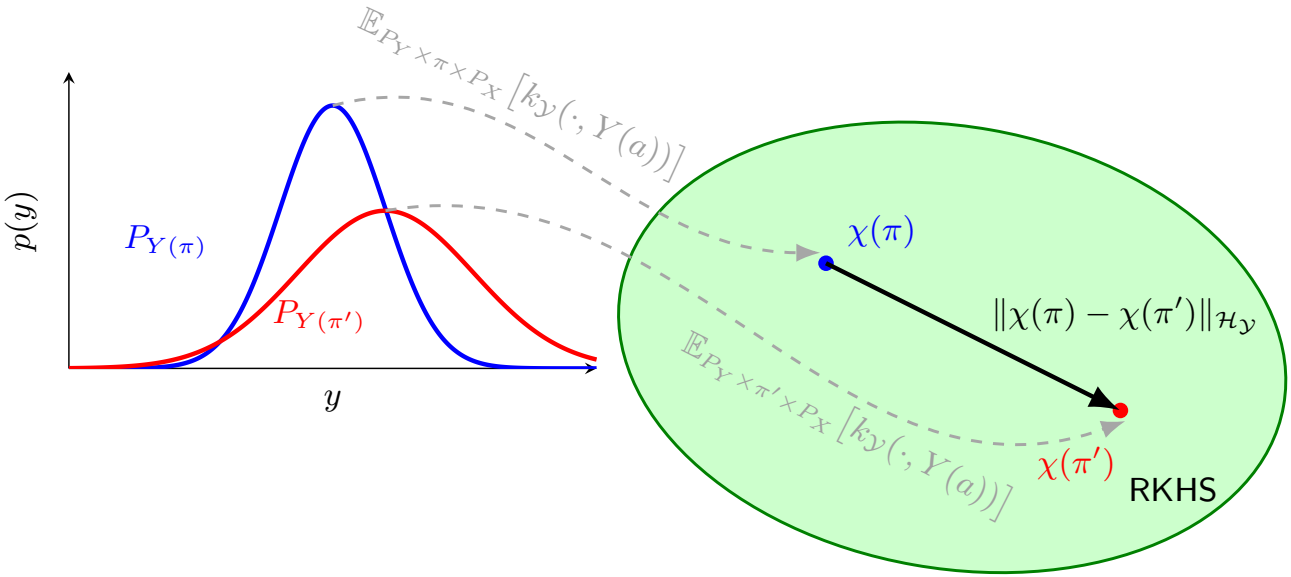
- Properties:
  - If k is *characteristic*, $\chi(\pi)$ uniquely identifies the distribution of $\{Y(a)\}_{a\sim\pi}$.
  - Rich, nonparametric representation of counterfactual distributions.
- **Assumptions (selection on observables).**
  1. Consistency / no interference: $Y = Y(a)$ when $A = a$.
  2. Conditional exchangeability: $Y(a) \perp A \mid X$.
  3. Strong positivity: $\inf_{a,x} \pi_0(a \mid x) > 0$ (essential infimum under P_X).
- Then, under the assumptions above, the **identified** counterfactual policy mean embedding writes:

$$\begin{aligned} \chi(\pi) &= \mathbb{E}_{\pi \times P_X}\big[\mu_{Y|A,X}(a,x)\big], \\ \mu_{Y|A,X}(a,x) &:= \mathbb{E}\big[k_{\mathcal{Y}}(\cdot, Y) \mid A = a,\ X = x\big] \in \mathcal{H}_{\mathcal{Y}}, \end{aligned}$$

[1] Our setting extends Counterfactual Mean Embeddings of Muandet et al. (2021) to off-policy evaluation.

Counterfactual Policy Mean Embedding

- **Intuition**
  - $\chi(\pi)$ is a representation of the entire distribution of $\{Y(a)\}_{a\sim\pi}$.
  - Each distribution becomes a point in the Hilbert space $\mathcal{H}_\mathcal{Y}$.
  - Distances in $\mathcal{H}_\mathcal{Y}$ correspond to differences between distributions.
  - $\|\chi(\pi) - \chi(\pi')\|_{\mathcal{H}_\mathcal{Y}}$ is zero iff distributions match (characteristic $k_\mathcal{Y}$); larger norms $\Rightarrow$ stronger distributional shift.



- **Why it is suited to structured outcomes:** choose $k_\mathcal{Y}$ to encode geometry of images/text/graphs (e.g., RBF on embeddings, string/graph kernels), then $\chi(\pi)$ summarizes the whole counterfactual distribution in that geometry.

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A Plug-in CPME estimator

- **Regularity (RKHS).** Assume bounded, continuous kernels $k_{\mathcal{AX}}$ and $k_{\mathcal{Y}}$; measurability; and $k_{\mathcal{Y}}$ characteristic.
- Define the **Conditional mean embedding**:

$$\mu_{Y|A,X}(a,x) \;:=\; \mathbb{E}\big[k_{\mathcal{Y}}(\cdot,Y) \mid A=a,\; X=x\big] \;\in\; \mathcal{H}_{\mathcal{Y}},$$

- **Conditional mean operator (CMO)**. The CME defines a bounded linear operator

$$C_{Y|A,X} : \mathcal{H}_{\mathcal{AX}} \;\to\; \mathcal{H}_{\mathcal{Y}}, \qquad C_{Y|A,X}\,\phi_{\mathcal{AX}}(a,x) \;=\; \mu_{Y|A,X}(a,x),$$

  where $\phi_{\mathcal{AX}}(a,x) = k_{\mathcal{AX}}(\cdot,(a,x))$ is the joint feature map. Hence, for any distribution over (A,X),

$$\mathbb{E}_{A,X}\left[\mu_{Y|A,X}(A,X)\right] \;=\; C_{Y|A,X}\;\mathbb{E}_{A,X}\left[\phi_{\mathcal{AX}}(A,X)\right].$$

- Then, when introducing the following kernel mean embedding:

$$\mu_{\pi} \;:=\; \mathbb{E}_X\;\mathbb{E}_{a\sim\pi(\cdot|X)}\big[\phi_{\mathcal{AX}}(a,X)\big] \;\in\; \mathcal{H}_{\mathcal{AX}},$$

- we obtain the following **decoupled formulation of CPME**

$$\boxed{\chi(\pi) \;=\; C_{Y|A,X}\,\mu_{\pi}} \in\; \mathcal{H}_{\mathcal{Y}}$$

Plug-in Estimator: closed form solutions

Goal. Estimate $C_{Y|A,X}$ and μ_π, then plug in:

$$\boxed{\hat{\chi}_{\mathrm{pi}}(\pi) \;=\; \hat{C}_{Y|A,X}\,\hat{\mu}_\pi} \;\in\; \mathcal{H}_\mathcal{Y}.$$

Estimate $\hat{C}_{Y|A,X}$ (operator ridge). For $\lambda > 0$,

$$\hat{\mathcal{L}}^c(C) \;=\; \frac{1}{n}\sum_{i=1}^{n} \left\| k_\mathcal{Y}(\cdot, y_i) \;-\; C\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a_i, x_i)) \right\|^2_{\mathcal{H}_\mathcal{Y}} + \lambda \| C \|^2_{HS}.$$

Let $\hat{C}_{Y|A,X} = \arg\min_C \hat{\mathcal{L}}^c(C)$. By kernelization,

$$\hat{\mu}_{Y|A,X}(a, x) \;=\; \hat{C}_{Y|A,X}\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x)) \;=\; \sum_{i=1}^{n} \beta_i(a, x)\, k_\mathcal{Y}(\cdot, y_i),$$

$$\beta(a, x) \;=\; \left(K_{AX,AX} + n\lambda I\right)^{-1} K_{AX,ax},$$

Estimate $\hat{\mu}_\pi$.

$$\text{Discrete } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\sum_{a\in\mathcal{A}} \pi(a \mid x_i)\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x_i)).$$

$$\text{Continuous } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{M}\sum_{m=1}^{M} k_{\mathcal{A}\mathcal{X}}(\cdot, (a_{im}, x_i)),\; a_{im} \sim \pi(\cdot \mid x_i).$$

Plug-in Consistency and Rates (SRC & EVD)

Proposition (Plug-in convergence rate in RKHS)

Under selection-on-observables, RKHS regularity, and standard source (smoothness $c \in (1, 3]$) and eigenvalue decay (complexity $b \in (0, 1]$) assumptions, choose

$$\lambda \;=\; n^{-1/(c+1/b)}.$$

Then, with high probability,

$$\left\|\hat{\chi}_{\mathrm{pi}}(\pi) - \chi(\pi)\right\|_{\mathcal{H}_{\mathcal{Y}}} = \mathcal{O}_p\Big(r_C(n)\Big) = \mathcal{O}_p\Big(n^{-\frac{c-1}{2(c+1/b)}}\Big)\ .$$

Interpretation.

- Error is dominated by estimating $C_{Y|A,X}$ (operator regression); $\hat{\mu}_\pi$ attains $n^{-1/2}$.
- b smaller $\Rightarrow$ slower spectral decay (harder problem); c larger $\Rightarrow$ smoother operator (easier).
- Best rate $= n^{-1/4}$ when $c = 3$ (favorable smoothness).

Teaser. A **doubly robust** estimator of CPME achieves root-n inference when nuisances are learned well (next section).

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The Efficient Influence Function (EIF) for the CPME

Goal. Construct an estimator of $\chi(\pi)$ that is *first-order unbiased* and *efficient*. A naïve plug-in $\hat{\chi}_{\mathrm{pi}}$ inherits bias from estimated nuisances ($\hat{\mu}_{Y|A,X}$, $\hat{\pi}_0$); the EIF provides the optimal correction.

EIF for the CPME (at the truth).

$$\psi^{\pi}(y, a, x) = \frac{\pi(a \mid x)}{\pi_0(a \mid x)} \{k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X}(a, x)\} + \int \mu_{Y|A,X}(a', x)\pi(da' \mid x) - \chi(\pi).$$

Intuition by components.

- $\frac{\pi}{\pi_0}(k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X})$: *IPW residual* — reweights the observed feature $k_{\mathcal{Y}}(\cdot, y)$ to emulate sampling under π, removing regression bias.
- $\int \mu_{Y|A,X}(a', x)\pi(da' \mid x)$: *Regression correction* — restores the target conditional mean under π.
- $-\chi(\pi)$: *Centering term* ensuring $\mathbb{E}[\psi^{\pi}] = 0$, so that empirical averages yield unbiased corrections.

The EIF yields a **doubly robust** estimator, consistent if $\mu_{Y|A,X}$ or π_0 is well-specified.

The Efficient Influence Function (EIF) for the CPME - II

Let $\widehat{P}$ denote the empirical law induced by the estimated nuisances $(\hat{\mu}_{Y|A,X},\, \hat{\pi}_0)$. The plug-in estimator $\widehat{\chi}_{\mathrm{pi}}(\pi) = \chi(\pi; \widehat{P})$ then inherits bias from these nuisance estimates.

By definition of EIF ψ^π acts as the *gradient* of the functional $\chi(\pi; P)$ with respect to the data-generating law P. Since $\chi(\pi; P)$ is pathwise differentiable, it admits a first-order (linear) expansion around the true distribution P_0:

$$\chi(\pi; P) - \chi(\pi; P_0) = \mathbb{E}_{P_0}\left[\psi^\pi(Z)\, \frac{dP - dP_0}{dP_0}(Z)\right] + o(\|P - P_0\|).$$

Replacing P_0 by $\widehat{P}$ gives

$$\chi(\pi; P_0) - \chi(\pi; \widehat{P}) \;\approx\; \mathbb{E}_{\widehat{P}}[\psi^\pi(Z)],$$

meaning that the EIF ψ^π captures how small estimation errors in the distribution $(\widehat{P} - P_0)$ translate into first-order changes in the parameter.

This expansion motivates the *one-step* or *doubly robust* estimator:

$$\widehat{\chi}_{\mathrm{dr}}(\pi) = \widehat{\chi}_{\mathrm{pi}}(\pi) + \tfrac{1}{n} \sum_{i=1}^{n} \widehat{\psi}^\pi(Z_i).$$

The EIF corrects the first order plug-in bias, yielding a **semiparametric efficient** estimator, achieving the minimum possible asymptotic variance.

From Plug-in to One-Step: Double Robustness and Better Rates

One-step DR estimator (using estimated nuisances):

$$\hat{\chi}_{\mathrm{dr}}(\pi) = \hat{\chi}_{\mathrm{pi}}(\pi) + \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\pi(a_i \mid x_i)}{\hat{\pi}_0(a_i \mid x_i)}\left\{k_{\mathcal{Y}}(\cdot, y_i) - \hat{\mu}_{Y|A,X}(a_i, x_i)\right\} + \int \hat{\mu}_{Y|A,X}(a, x_i)\pi(da' \mid x_i) - \hat{\chi}_{\mathrm{pi}}(\pi)\right].$$

Why it's better than plug-in:

Proposition (DR convergence rate in RKHS)

- **Double robustness.** *Consistent if either $\hat{\pi}_0$ or $\hat{\mu}_{Y|A,X}$ is correct.*
- **Rate improvement.** *In $\mathcal{H}_{\mathcal{Y}}$,*

$$\left\|\hat{\chi}_{\mathrm{dr}}(\pi) - \chi(\pi)\right\| = \mathcal{O}_p\Big(n^{-1/2} + r_{\pi_0}(n)\, r_C(n)\Big).$$

  If $r_{\pi_0}(n) \asymp r_C(n) \asymp n^{-1/4}$, the product is $n^{-1/2} \Rightarrow \sqrt{n}$-rate—improving over the plug-in's $n^{-1/4}$ ceiling.
- **Bias cancellation.** *The IPW residual term removes first-order plug-in bias; the augmentation stabilizes variance and reduces model misspecification error.*
- **Efficiency.** *The EIF achieves the semiparametric efficiency bound (minimal asymptotic variance among regular estimators) under standard conditions.*

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Application I: Testing with CPME, Distributional OPE

Goal. Compare counterfactual outcome distributions under two policies π, π':

$$H_0 : \mathbb{E}_{P_\pi}\left[k_{\mathcal{Y}}(\cdot, Y)\right] = \mathbb{E}_{P_{\pi'}}\left[k_{\mathcal{Y}}(\cdot, Y)\right].$$

EIF-based moment (difference). Let

$$\mu_{Y|A,X}(a,x) := \mathbb{E}\big[k_{\mathcal{Y}}(\cdot, Y) \mid A = a, X = x\big], \qquad \beta_\pi(x) := \mathbb{E}_{a\sim\pi(\cdot|x)}[\mu_{Y|A,X}(a,x)].$$

Define the Hilbert-valued score

$$\varphi_{\pi,\pi'}(y,a,x) = \left\{\frac{\pi(a|x)}{\pi_0(a|x)} - \frac{\pi'(a|x)}{\pi_0(a|x)}\right\}\Big\{k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X}(a,x)\Big\} + \beta_\pi(x) - \beta_{\pi'}(x).$$

Then

$$H_0 \quad \Longleftrightarrow \quad \mathbb{E}[\varphi_{\pi,\pi'}(Y,A,X)] = 0.$$

Why EIF?

- *Doubly robust:* unbiased if either π_0 or $\mu_{Y|A,X}$ is correctly specified.
- *Orthogonal moment:* locally insensitive to nuisance errors, yielding higher power with learned nuisances.

Motivation. Before deployment, test—using logged data—whether a new policy π significantly alters the *entire* outcome distribution compared to the baseline π_0.

Application I: DR-KPT - cross U-statistic and decision rule

Split & estimate. Split the sample: first $m = \lfloor n/2 \rfloor$, last $n - m$. Fit $\hat{\pi}_0, \hat{\mu}_{Y|A,X}$ on each split separately and compute $\hat{\varphi}$ on the first split, $\tilde{\varphi}$ on the second.
Cross U statistic. For $i \leq m$,

$$f^{\dagger}_{\pi,\pi'}(Y_i, A_i, X_i) = \frac{1}{n-m} \sum_{j=m+1}^{n} \left\langle \hat{\varphi}_{\pi,\pi'}(Y_i, A_i, X_i),\, \tilde{\varphi}_{\pi,\pi'}(Y_j, A_j, X_j) \right\rangle_{\mathcal{H}_{\mathcal{Y}}}.$$

Let

$$\bar{f}^{\dagger} = \frac{1}{m} \sum_{i=1}^{m} f^{\dagger}_{\pi,\pi'}(Y_i, A_i, X_i), \qquad S^{\dagger} = \sqrt{\frac{1}{m} \sum_{i=1}^{m} \left(f^{\dagger}_{\pi,\pi'} - \bar{f}^{\dagger}\right)^2},$$

$$\boxed{T^{\dagger}_{\pi,\pi'} = \frac{\sqrt{m}\, \bar{f}^{\dagger}}{S^{\dagger}}}.$$

Proposition (Asymptotic normality & decision.)

Under regularity and $r_{\pi_0}(n)\, r_C(n) = \mathcal{O}(n^{-1/2})$, $T^{\dagger}_{\pi,\pi'} \Rightarrow \mathcal{N}(0,1)$. *Compute* $p = 1 - \Phi(T^{\dagger}_{\pi,\pi'})$; *reject* H_0 *at level* α *if* $p \leq \alpha$.

Notes. Cross U handles MMD degeneracy; $m = \lfloor n/2 \rfloor$ maximizes power; EIF-based (DR) estimation boosts rate and sensitivity versus pure plug-in or IPW.

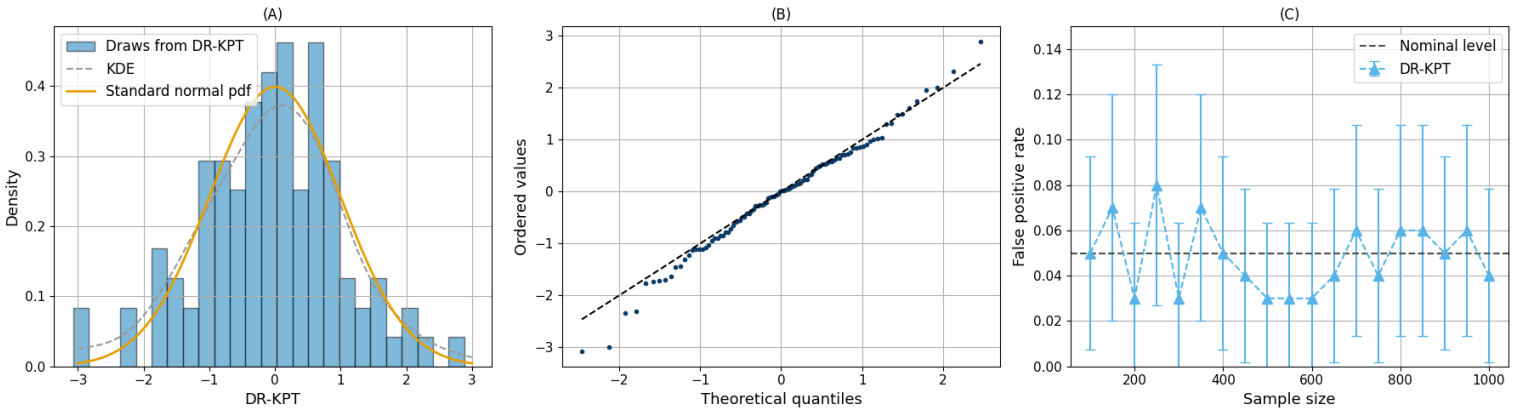
Application I: DR-KPT - experimental setup & calibration

Data. i.i.d. $(X_i, A_i, Y_i)_{i=1}^n$; continuous actions; unknown nuisances.
Nuisance learning. $\hat{\pi}_0(\cdot \mid x)$: linear regression; $\hat{\mu}_{Y|A,X}$: conditional mean embeddings.
Level. All tests at $\alpha = 0.05$. 100 simulations.

Scenarios.

- *I (Null)*: $\pi = \pi' \Rightarrow P_{Y(\pi)} = P_{Y(\pi')}$.
- *II (Mean shift)*: $P_{Y(\pi)}, P_{Y(\pi')}$ differ in means.
- *III (Mixture)*: $P_{Y(\pi')}$ is a mixture; mean unchanged compared to $P_{Y(\pi)} \Rightarrow$ bimodal shift.
- *IV (Shifted mixture)*: mixture + mean shift.

Calibration under the null (Scenario I). DR-KPT statistic $T^{\dagger}_{\pi,\pi'}$ is approximately $\mathcal{N}(0,1)$ and the empirical type-I error matches 0.05.



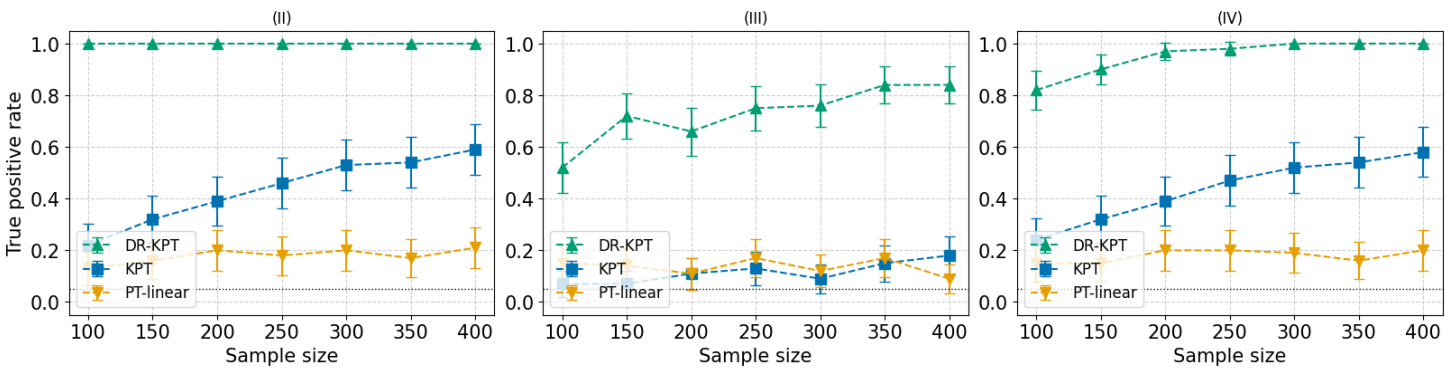
DR-KPT vs. baselines: power under Scenarios II–IV

Baselines.

- **KPT** (adapted from KTE, *estimated* propensities).
- **PT-linear**: linear kernel (tests only mean differences).

Result. DR-KPT shows higher true-positive rates in all alternatives:

- Detects pure mean shifts (Scenario II) and non-mean shifts (bimodality, Scen. III) as well as combined shifts (Scen. IV).
- KPT / PT-linear often lose power due to propensity estimation error or insensitivity to higher moments.
- EIF-based, doubly robust construction $\Rightarrow$ better calibration + power with learned nuisances; cross-U handles MMD degeneracy; computation is efficient.



Application I: Runtime — DR-KPT is *orders of magnitude* faster

Summary (Scenarios I–IV). DR-KPT achieves the best computational efficiency—typically $\mathbf{10^2}\times$ faster— thanks to its *closed-form* test statistic, avoiding costly resampling or kernel-matrix permutations.

Why: KPT relies on permutation-based MMD with repeated kernel recomputation, while PT-linear still requires multiple reweighting steps. In contrast, DR-KPT computes a single analytic statistic with no resampling.

Table: Average runtime (s) for Scenario I (mean $\pm$ std over 100 runs).

Sample size	KPT	PT-linear	DR-KPT
100	0.495 $\pm$ 0.070	0.592 $\pm$ 0.061	**0.004 $\pm$ 0.005**
150	0.740 $\pm$ 0.039	0.774 $\pm$ 0.038	**0.007 $\pm$ 0.004**
200	1.134 $\pm$ 0.081	1.060 $\pm$ 0.051	**0.010 $\pm$ 0.009**
250	1.623 $\pm$ 0.075	1.553 $\pm$ 0.076	**0.008 $\pm$ 0.002**
300	2.257 $\pm$ 0.074	2.373 $\pm$ 0.202	**0.013 $\pm$ 0.007**
350	3.204 $\pm$ 0.118	3.384 $\pm$ 0.160	**0.025 $\pm$ 0.023**
400	4.180 $\pm$ 0.136	4.358 $\pm$ 0.251	**0.019 $\pm$ 0.007**

DR-KPT's closed-form expression removes the need for kernel recomputation or randomization, making it up to two orders of magnitude faster than classical kernel tests.

Application II - Sampling from the counterfactual distribution

Goal. Use the estimated embedding $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ to generate pseudo-samples $\tilde{y}_{1:m}$ whose empirical law approximates the outcome distribution under π.

Deterministic kernel herding (from $\hat{\chi}(\pi)$).

$$\tilde{y}_1 := \arg\max_{y \in \mathcal{Y}} \hat{\chi}(\pi)(y), \qquad \tilde{y}_t := \arg\max_{y \in \mathcal{Y}} \left[\hat{\chi}(\pi)(y) - \tfrac{1}{t} \sum_{\ell=1}^{t-1} k_{\mathcal{Y}}(\tilde{y}_\ell, y)\right], \; t = 2, \ldots, m.$$

Proposition (Guarantees (MMD & weak convergence))

Under mild regularity (locally compact $\mathcal{Y}$, $\mathcal{H}_{\mathcal{Y}} \subset \mathcal{C}_0$):

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,pi}, \nu(\pi)\right) = \mathcal{O}_p\big(r_C(n, b, c) + m^{-1/2}\big),$$

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,dr}, \nu(\pi)\right) = \mathcal{O}_p\big(n^{-1/2} + r_{\pi_0}(n) r_C(n, b, c) + m^{-1/2}\big),$$

$\Rightarrow \tilde{y}_{pi,j} \rightsquigarrow \nu(\pi)$ *and* $\tilde{y}_{dr,j} \rightsquigarrow \nu(\pi)$.

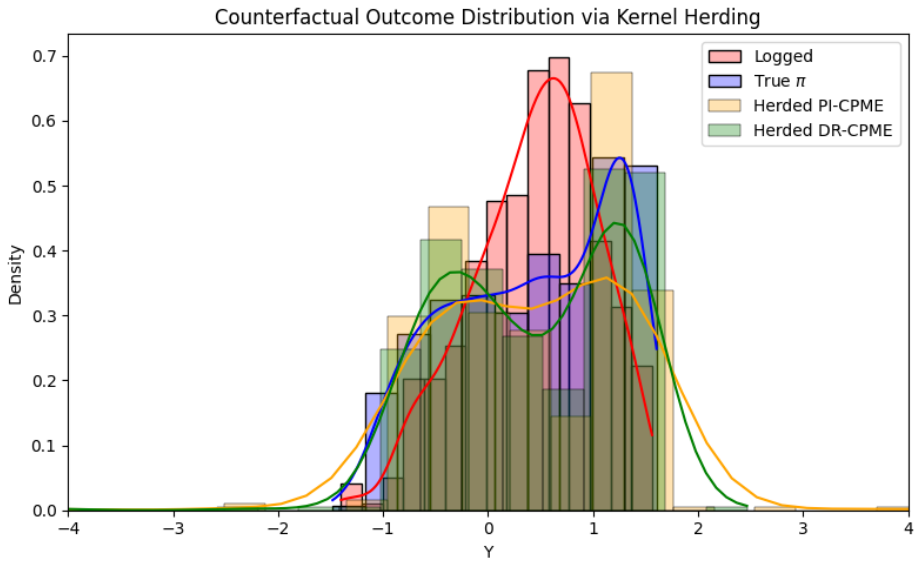
m trades compute for quality; DR embedding improves sample quality via the faster $n^{-1/2}$-driven term.

Motivation. In simulation studies, we can arbitrarily sample from any given policy to assess the behavior of our test under controlled conditions—allowing us to validate its sensitivity to distributional shifts.

Application II: Sampling via kernel herding - setup & qualitative result

Goal. Generate counterfactual samples from $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ using kernel herding; compare **Plug-in** vs. **DR** CPME.
Scenarios. Logging policy $\pi_0 \in \{\text{uniform, logistic}\}$; outcome $g \in \{\text{quadratic, sinusoidal (nonlinear)}\}$.

- Herd from $\hat{\chi}_{\mathrm{pi}}(\pi)$ and $\hat{\chi}_{\mathrm{dr}}(\pi)$ (Alg. sampling).
- Compare herded empirical law $\tilde{P}_Y^m$ to oracle $\nu(\pi)$ (synthetic ground truth).
- Metrics: Wasserstein distance and MMD (Gaussian kernel).



Logistic π_0, nonlinear outcome: logged vs. oracle vs. Plug-in/DR herded.

Application II: Sampling fidelity - Wasserstein & MMD (herded vs. oracle)

Wasserstein distance

Method	log–nonlin	log–quad	uni–nonlin	uni–quad
Plug-in	1.29e-01 ± 2.6e-01	1.41e-01 ± 4.9e-02	9.08e-02 ± 3.7e-01	6.78e-02 ± 1.9e-02
DR	**8.60e-02** ± 2.2e-02	1.36e-01 ± 3.9e-02	**5.00e-02** ± 1.5e-02	**6.63e-02** ± 1.6e-02

MMD (Gaussian kernel)

Method	log–nonlin	log–quad	uni–nonlin	uni–quad
Plug-in	1.11e-03 ± 5.9e-03	9.85e-04 ± 6.0e-04	1.92e-04 ± 1.2e-03	3.31e-04 ± 2.5e-04
DR	**4.38e-04** ± 3.6e-04	9.80e-04 ± 6.0e-04	**6.49e-05** ± 4.4e-05	3.51e-04 ± 2.5e-04

Takeaways.

- DR herding matches oracle better (lower Wasserstein/MMD), especially under *nonlinear* outcomes and *logistic* logging where misspecification is hardest.
- Consistent with theory: DR CPME improves the embedding ($n^{-1/2}$-driven term), yielding higher-fidelity samples from $\nu(\pi)$.

Discussion

Summary. We introduced the **Counterfactual Policy Mean Embedding (CPME)** (Zenati et al., 2025) — a general kernel-based framework for *distributional off-policy evaluation*. It unifies plug-in and **EIF-based doubly robust** estimation, enables the **DR-KPT** test (a closed-form, non-permutation kernel test), and supports deterministic counterfactual sampling via kernel herding.

Limitations.

- Standard identification assumptions: consistency, exchangeability, and positivity.
- RKHS regularity and kernel choice; computational scalability of kernel operations ($O(n^2)$).

Future work.

- *Distributional policy optimization*: learn policies by optimizing CPMEs under risk or safety constraints.
- *Robustness*: develop sensitivity analyses, proximal or instrumental-variable extensions, and partial identification bounds for $\chi(\pi)$.

Ongoing direction: extending kernel treatment effects to **adaptively collected data**. New preprint and code: `https://github.com/houssamzenati/adaptive-KTE`

References

 Muandet, Krikamol et al. (2021). "Counterfactual mean embeddings". In: *Journal of Machine Learning Research* 22.162, pp. 1–71.

 Zenati, Houssam et al. (2025). "Doubly-Robust Estimation of Counterfactual Policy Mean Embeddings". In: *Neural Information Processing Systems*.

The Efficient Influence Function (EIF) for the CPME – I

Pathwise view. Consider a regular submodel $\{P_\varepsilon : \varepsilon \in \mathbb{R}\}$ through the true law P_0, with density

$$p_\varepsilon(z) = p_0(z)\{1 + \varepsilon s(z)\} + o(\varepsilon),$$

where $s(Z) \in L^2(P_0)$ is the **score function**. It encodes a valid infinitesimal perturbation of P_0: the directions in which the data law can move smoothly.

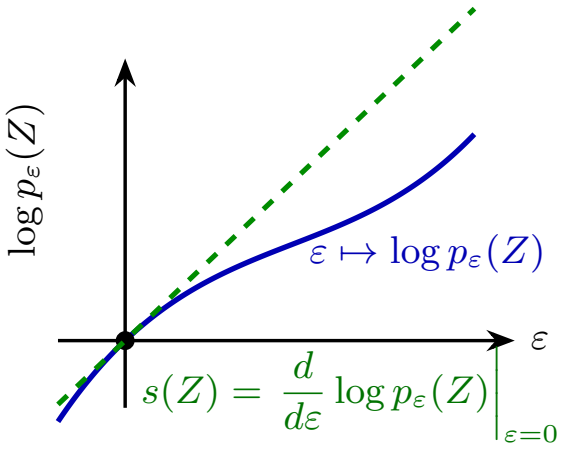
The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi) \in \mathcal{H}_\mathcal{Y}$ is **pathwise differentiable** if for every score s,

$$D_s\chi(\pi) := \left.\frac{d}{d\varepsilon}\chi_\varepsilon(\pi)\right|_0 \text{ exists in } \mathcal{H}_\mathcal{Y}.$$

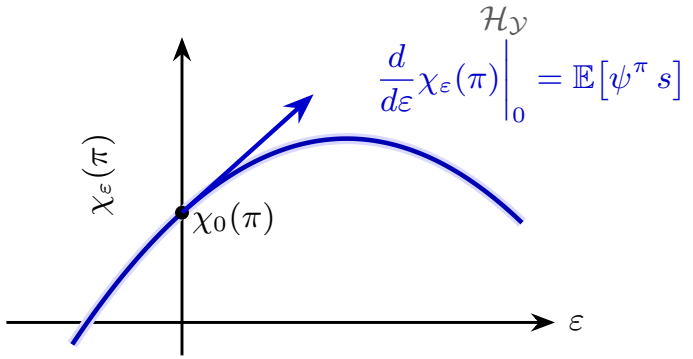
By the Riesz representation theorem, there exists a unique $\psi^\pi(Y, A, X) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ such that

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s],$$

defining the **efficient influence function (EIF)** as the Riesz representer of the pathwise derivative.



Left: $s(Z) = \partial_\varepsilon \log p_\varepsilon(Z)|_0$ is the slope of the log-density; it specifies the direction of infinitesimal change in P_ε.



Right: The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi)$ in $\mathcal{H}_\mathcal{Y}$; its tangent at 0 is $\mathbb{E}[\psi^\pi s]$, defining the EIF as the functional "gradient" of $\chi(\pi)$ with respect to the data law.

The Efficient Influence Function (EIF) for the CPME – II

Influence functions. An *influence function* $h(Z) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ characterizes how a parameter $\chi(P)$ reacts to infinitesimal perturbations of the data law:

$$\frac{d}{d\varepsilon}\chi(P_\varepsilon)\Big|_0 = \mathbb{E}_{P_0}[h(Z)s(Z)], \qquad \mathbb{E}_{P_0}[h(Z)] = 0.$$

It quantifies how each observation Z contributes to the change in $\chi(P)$ when P is locally perturbed along the *score direction* $s(Z)$.

Why the tangent space. Let $\mathcal{T} \subseteq L^2(P_0)$ denote the **tangent space**—the closure of all valid score functions. It represents the set of directions in which the distribution P_0 can vary smoothly within the model. Projecting onto $\mathcal{T}$ ensures that we only consider directions that correspond to *real, estimable* perturbations of the data law.

Geometric interpretation. The **EIF** $\psi^\pi \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ is the *unique element* whose inner product with any score reproduces the pathwise derivative:

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s], \quad \forall s \in \mathcal{T}.$$

Equivalently, ψ^π is the *orthogonal projection* of any regular influence function h onto the tangent space $\mathcal{T}$—it isolates the part of h that corresponds to meaningful, model-consistent variation in P.

The Efficient Influence Function (EIF) for the CPME – III

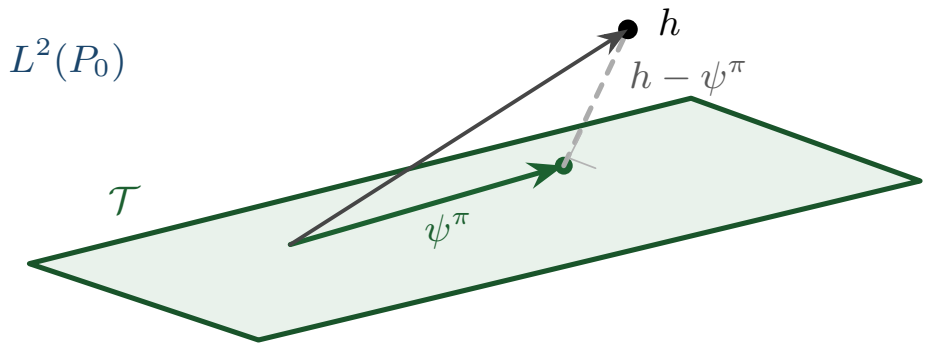


Figure: The EIF ψ^π is the orthogonal projection of any regular influence function h onto the tangent space $\mathcal{T} \subset L^2(P_0)$. The residual $h - \psi^\pi$ is orthogonal to $\mathcal{T}$, meaning that it corresponds to directions that cannot change $\chi(P)$.

Why it minimizes variance. Since $h - \psi^\pi$ is orthogonal to all valid scores:

$$\mathbb{E}_{P_0}[(h - \psi^\pi)s] = 0, \quad \forall s \in \mathcal{T},$$

the EIF ψ^π is the projection of h onto $\mathcal{T}$. By the Pythagorean theorem in $L^2(P_0)$,

$$\|h\|^2_{L^2(P_0)} = \|\psi^\pi\|^2_{L^2(P_0)} + \|h - \psi^\pi\|^2_{L^2(P_0)} \Rightarrow \|h\|^2_{L^2(P_0)} \geq \|\psi^\pi\|^2_{L^2(P_0)}.$$

Thus, the EIF yields the **smallest possible asymptotic variance**: it is the most efficient (shortest) direction in $L^2(P_0)$ that still preserves the correct local behavior of $\chi(\pi)$ under valid perturbations of P.